C PROGRAMMING FROM SCRATCH · EPISODE 04

The main Function

Where Every C Program Starts

KEY POINTS

- Execution always begins at main and runs from the top of main downward.
- One program, one main, even when the project has several .c files.
- int promises an integer and return 0 delivers it. 0 means a normal exit; non-zero, an abnormal one.
- Header = return type, name, parameters. Body = the curly braces. Every statement goes inside.

01 Where Execution Starts

▶ VIDEO 01:09

Statements in main run top to bottom, one after another.

three-prints.c — C SOURCE

```c
#include <stdio.h>

int main()
{
    printf("hello C\n");
    printf("hello C\n");
    printf("hello C\n");
    return 0;
}
```

Microsoft Visual Studio Debug Console

```
hello C
hello C
hello C

...\x64\Debug\first_program.exe exited with code 0.
Press any key to close this window . . .
```

Debug › Step Over (F10) runs one statement per press. The first press stops at main's opening brace, not at `#include`. The yellow arrow marks the line that runs *next*.

02 One and Only One

▶ VIDEO 03:01

A second `int main()` in hello.c fails the build: *function 'int main()' already has a body*.

One project, one main: a main in a second file (second.c) fails the build too.

03 int and return 0

▶ VIDEO 04:23

- `int` (integer): main hands back a whole number when it finishes.
- `return 0;` ends main and returns 0. Visual Studio shows it as *exited with code 0*.

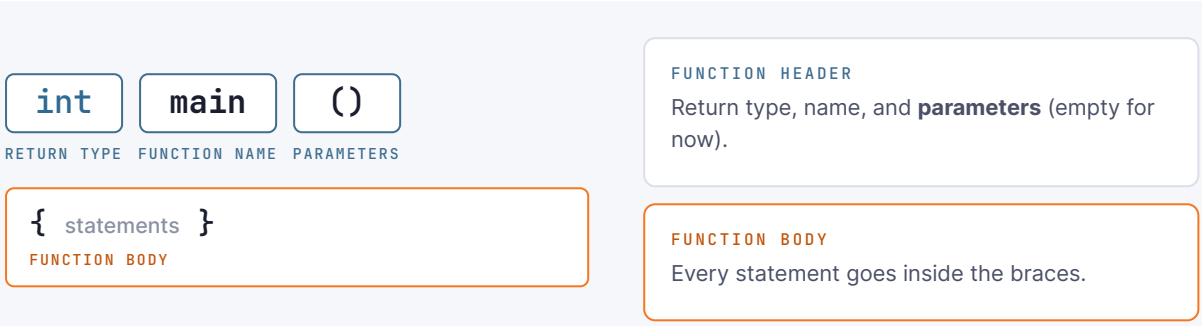
KEY IDEA

0 means the program finished normally; any non-zero value signals an abnormal exit.

04 Header and Body

▶ VIDEO 05:50

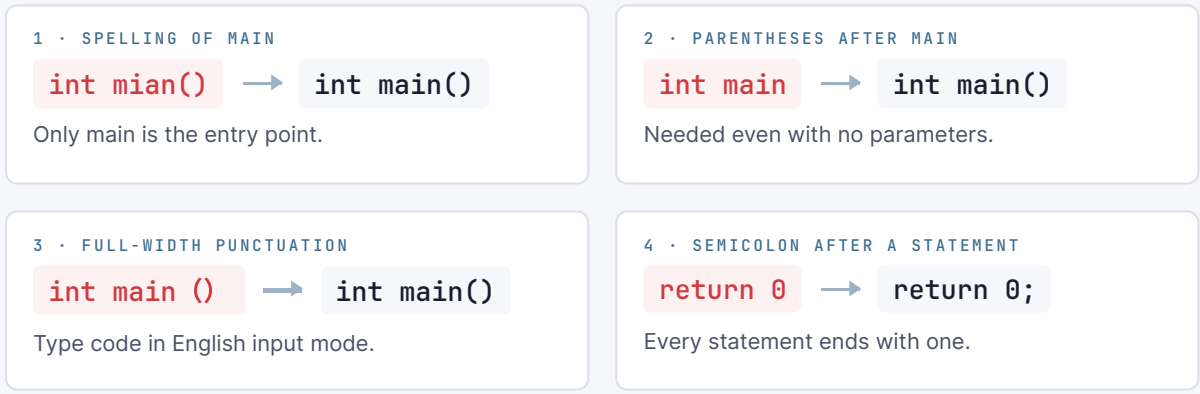
FIGURE 4.1 The anatomy of main



05 Four Common Mistakes

▶ VIDEO 07:20

FIGURE 4.2 Four slips and their fixes



06 A Template to Keep

▶ VIDEO 08:56

Type this skeleton first, then write your statements above `return 0;`.

template.c — A TEMPLATE TO KEEP

```c
int main()
{
    return 0;
}
```

CODE 11 files on GitHub github.com/abitclearer/c-programming-from-scratch/tree/main/04-the-main-function